Alcohol Label Verification

README — setup, structure, and how to run it locally

Live demo: https://alcohol-label-verification-fawn.vercel.app

Repository: github.com/aaryannaz/alcohol-label-verification

Summary

A web app that automates what a TTB reviewer does by hand on a COLA (Certificate of Label Approval) application: it reads the regulated fields off the label artwork, compares them to the approved application, and flags anything that does not match. The reviewer uploads the label image(s); Google Gemini Vision reads them into structured fields and pre-fills an editable set of COLA application fields; the reviewer corrects those to the approved application; a pure, rule-based engine then compares them field-by-field and runs the TTB compliance checks (required fields per beverage type, the exact government-warning text, additive disclosures, and more). Typical extraction is about 2 seconds; a batch mode processes many labels in parallel, each landing on a Pass / Needs-attention verdict.

Backend: FastAPI (Python 3.12). AI: Google Gemini Vision (gemini-2.5-flash). Frontend: vanilla HTML/CSS/JS served by the same app. Deployed on Vercel.

File structure

```
alcohol-label-verification/
  README.md  APPROACH.md  LICENSE        docs (this PDF mirrors README.md)
  requirements.txt                      runtime deps (installed by Vercel)
  pyproject.toml                        packaging + ruff (lint) config
  vercel.json   .vercelignore           deploy / routing config
  .python-version   .env.example       Python 3.12; copy to .env
  .github/workflows/ci.yml             CI: ruff + tests on every push
  app/
    main.py            FastAPI app: routes, exception handlers, middleware
    fields.py          canonical regulated-field list (single source of truth)
    schemas.py         Pydantic request models + product / origin enums
    clients.py         Gemini client + model / timeout config
    extraction.py      image -> JSON field extraction (Gemini, category-aware)
    prompts.py         the extraction prompts (most domain rules live here)
    validation.py      rule-based field comparison + TTB compliance logic
    uploads.py         upload validation (extension / content-type / signature)
    security.py        rate limiting, optional auth, security headers
    observability.py   structured logging + per-request correlation IDs
    errors.py          one JSON error envelope for all failures
    static/            browser UI (HTML/CSS/JS) served by FastAPI
  evals/               extraction-accuracy harness (render, score, cases)
  tests/               unit + API tests (test_validation.py, test_api.py)
  scripts/             gemini_smoke_test.py (Gemini connectivity check)
```

Run it locally (step by step)

1. Prerequisites. Python 3.12 and a Google Gemini API key (free from Google AI Studio: aistudio.google.com/apikey).

2. Get the code.

```
git clone https://github.com/aaryannaz/alcohol-label-verification.git
cd alcohol-label-verification
```

3. Set your API key. Copy the example env file and add your key.

```
cp .env.example .env
# then edit .env and set GEMINI_API_KEY=your-key
```

4. Create a virtualenv and install dependencies.

```
python -m venv venv
venv/bin/python -m pip install -r requirements.txt
```

5. Run the server (from the repository root).

```
venv/bin/python -m uvicorn app.main:app --reload
```

6. Open the app. UI at http://localhost:8000/ · API docs at http://localhost:8000/docs

7. (Optional) Run the tests and the accuracy eval.

```
venv/bin/python -m unittest discover tests          # unit + API tests
venv/bin/python -m evals.run_eval                   # extraction accuracy + latency
```